

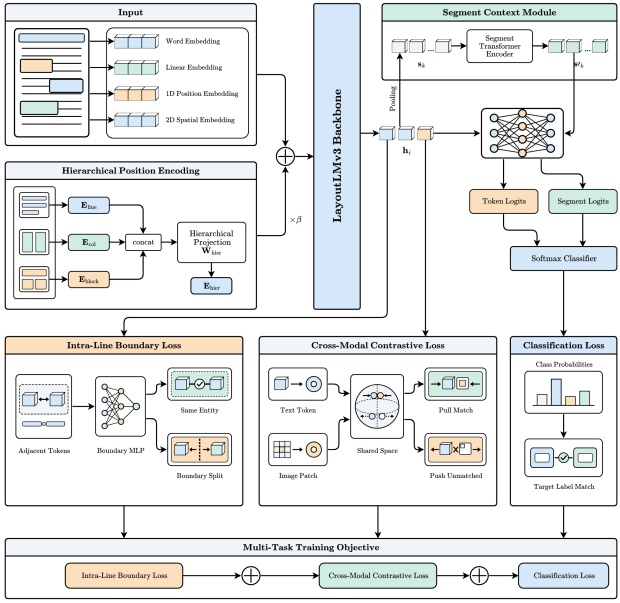
UNIVERSITY OF ENGINEERING AND TECHNOLOGY
VIETNAM NATIONAL UNIVERSITY HANOI

Weekly report

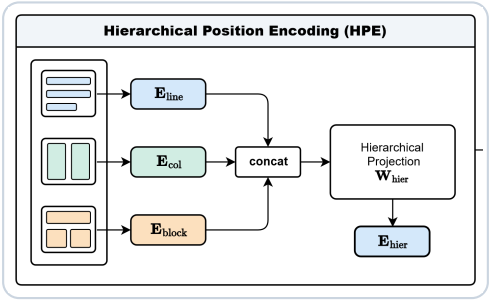
Presenter
Pham Nhat Minh
Major: Information Systems

Supervisor
PhD. Pham Thi Quynh Trang
Faculty of Information Technology, VNU-UET

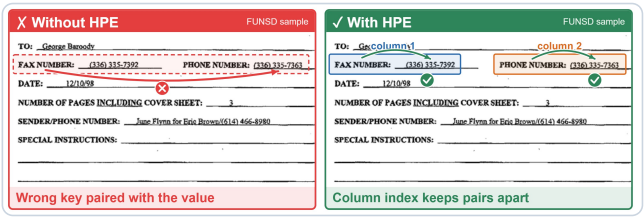
Model Architecture



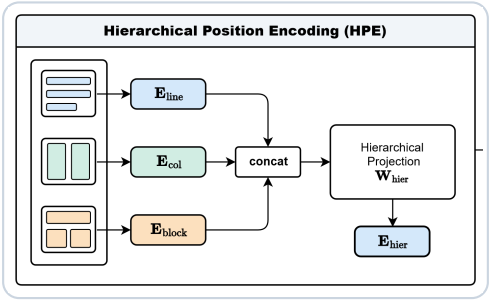
Hierarchical Position Encoding (HPE)



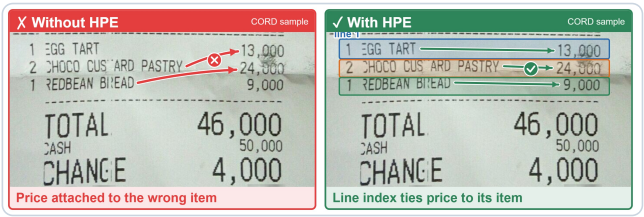
- **Internal Model Mechanics:** Assigns line, column, and block structural IDs to each token, projecting them into embeddings via a learnable scalar gate.
- **Real Document Impact (FUNSD Multi-Column):** Prevents cross-column attention confusion on multi-column tables, keeping keys paired with correct column values.



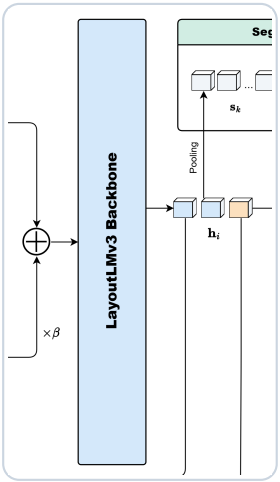
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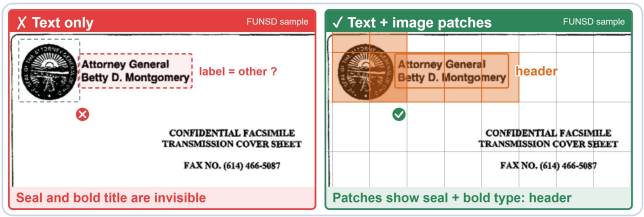
- **Internal Model Mechanics:** Encodes line sequence indices into token embeddings, preserving reading order and local grouping across stacked horizontal items.
- **Real Document Impact (CORD Multi-Row):** Without line indices, prices get bound to items on neighboring rows; HPE line coordinates tie each price strictly to its item.



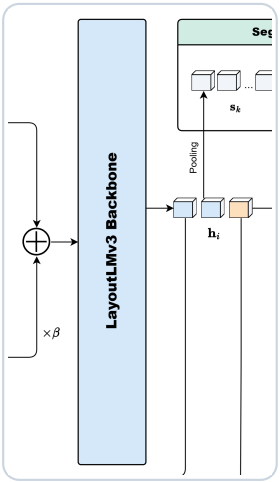
LayoutLMv3 Multimodal Backbone



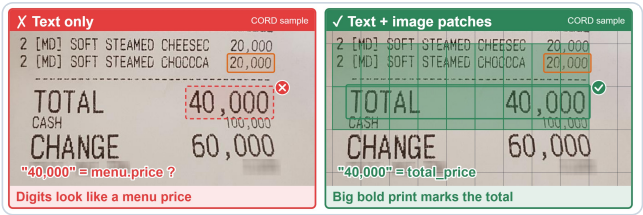
- **Internal Model Mechanics:** Jointly contextualizes subword text tokens and visual patches in shared self-attention with 2D relative spatial position bias.
- **Real Document Impact (FUNSD Header):** Visual patches recover critical entity semantics (headers, seals, logos) even when text OCR is noisy or missing.



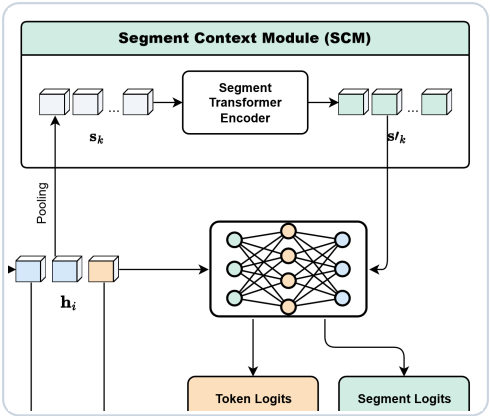
LayoutLMv3 Multimodal Backbone



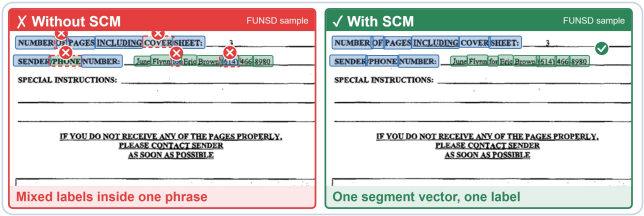
- **Internal Model Mechanics:** Integrates patch visual features (font size, weight, prominence) to distinguish visually distinct but lexically similar numeric strings.
- **Real Document Impact (CORD Grand Total):** Pure text sees digits as standard menu prices; visual patches capture big bold print to identify the true grand total.



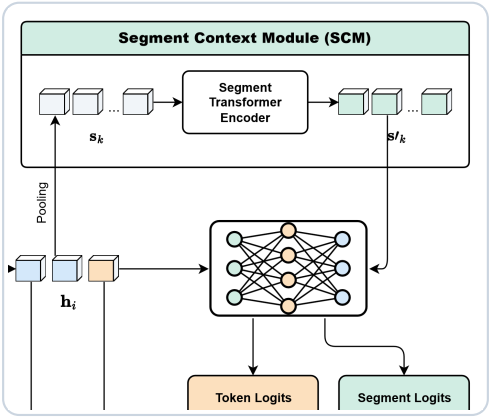
Segment Context Module (SCM)



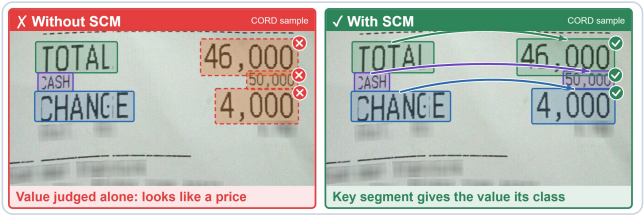
- **Internal Model Mechanics:** Pools token states into line-level segment vectors, propagates page-wide context, and injects segment consensus via gated residual fusion.
- **Real Document Impact (FUNSD Word Jitter):** Eliminates word-level label jitter within multi-token phrases, ensuring all tokens in a line share a unified semantic class.



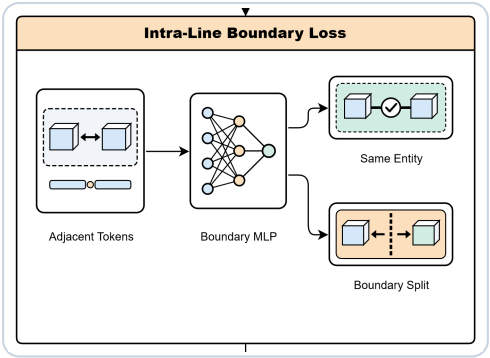
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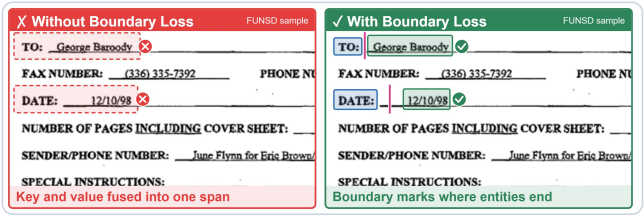
- **Internal Model Mechanics:** Propagates segment representations across lines via bidirectional attention, passing semantic context from keys to values.
- **Real Document Impact (CORD Value Binding):** Standalone numbers lack context on their own; SCM transfers the label from key segments (Total, Cash, Change) to classify values.



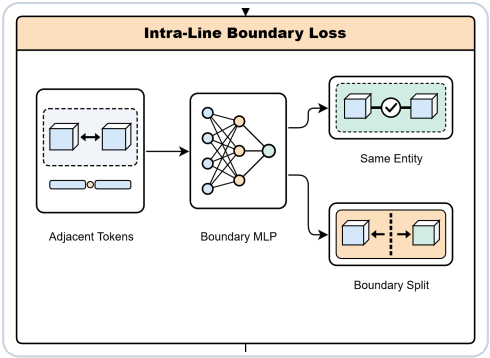
Intra-Line Boundary Loss



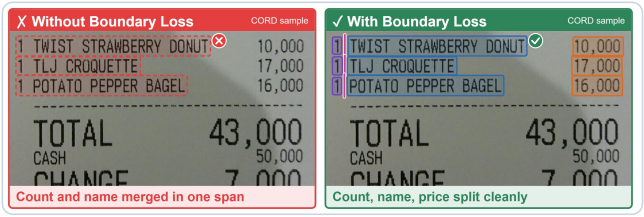
- **Internal Model Mechanics:** Supervises adjacent token pairs with a lightweight MLP transition classifier during training, adding zero latency at inference.
- **Real Document Impact (FUNSD Inline Field):** Prevents label bleed on inline fields (e.g. `TO: George Baroody`) by enforcing sharp transition cuts between keys and values.



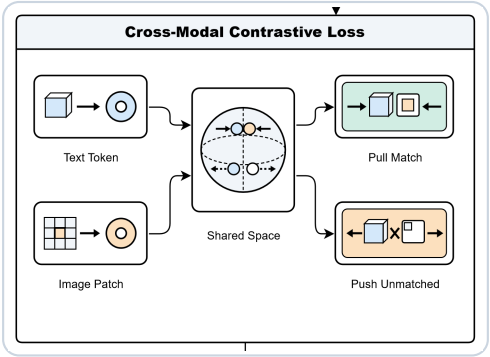
Intra-Line Boundary Loss



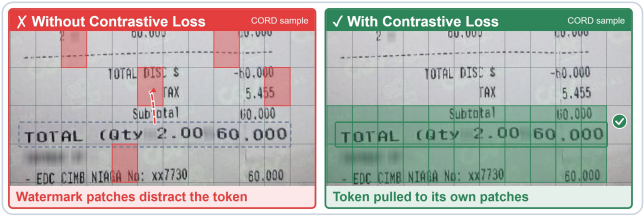
- **Internal Model Mechanics:** Detects entity shifts between consecutive horizontal tokens sharing the same line, penalizing blurred span transitions.
- **Real Document Impact (CORD Dense Item Row):** Dense receipt rows with count, name, and price merge without boundary supervision; boundary loss cleanly splits all three.



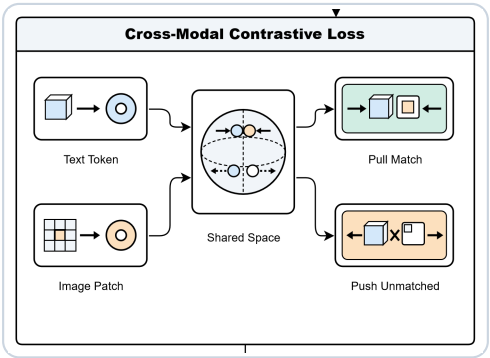
Cross-Modal Contrastive Loss



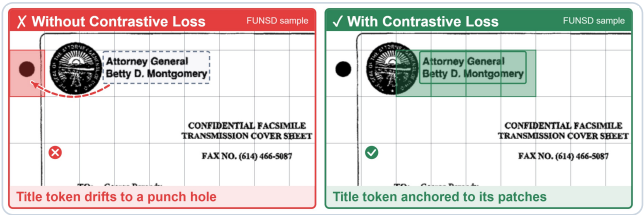
- **Internal Model Mechanics:** Aligns pre-SCM text projections with detached visual patch anchors via InfoNCE loss, preventing gradient interference.
- **Real Document Impact (CORD Watermark Noise):** Repels background watermark patterns that distract token attention, pulling text tokens firmly to their actual visual patches.



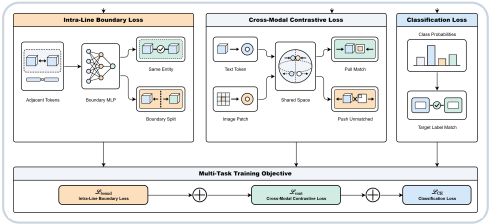
Cross-Modal Contrastive Loss



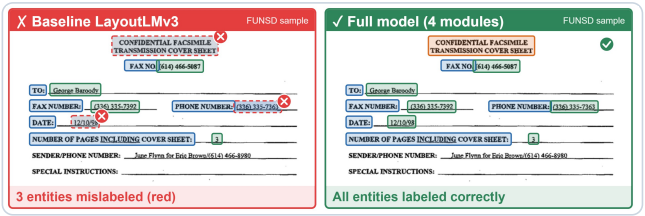
- **Internal Model Mechanics:** Soft-target spatial overlap pushes non-overlapping visual noise away from text embeddings, stabilizing multimodal representation learning.
- **Real Document Impact (FUNSD Margin Artifacts):** Prevents header tokens from drifting toward dark hole-punch margins or scanning borders, anchoring text to true print areas.



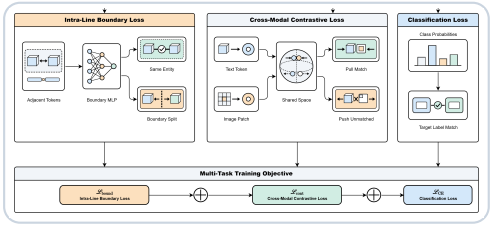
Multi-Task Training Objective



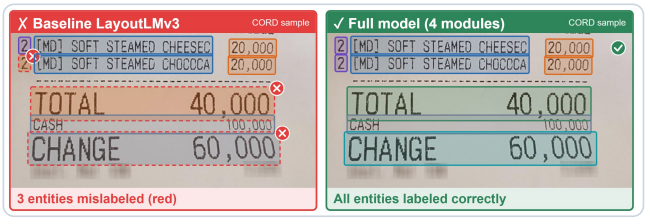
- **Internal Model Mechanics:** Jointly optimizes entity classification, boundary detection, and cross-modal contrastive alignment in a single unified backpropagation pass.
- **Real Document Impact (FUNSD Form):** Auxiliary structural tasks regularize the representation, eliminating entity misclassifications across complex document layouts.



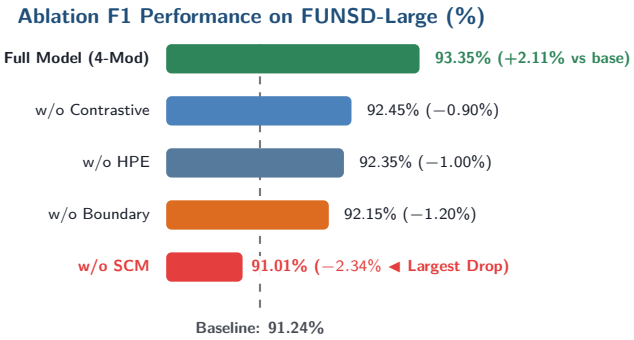
Multi-Task Training Objective



- **Internal Model Mechanics:** Combines multi-task gradients to ensure both fine-grained token boundaries and global visual-text alignment are jointly preserved.
- **Real Document Impact (CORD Receipt):** Resolves mislabeled total, count, and change fields seen in the baseline, achieving 100% field precision on complex receipts.



Training Results: Module Impact on F1 Score



- **SCM Has Greatest Impact:** Removing SCM triggers the largest F1 drop across all settings (-2.34% on FUNSD-Large, -1.79% on Base), confirming that segment-level context is the primary performance driver.
- **All Modules Improve Baseline:** Every single module configuration beats the baseline ($+0.44\% \cdots + 1.45\%$), with the Full Model achieving $+\mathbf{2.11}\%$ gain over standard LayoutLMv3.
- **Boundary & Contrastive Gains:** Intra-line boundary supervision adds $+1.20\%$ F1, while contrastive tuning adds $+0.90\%$ F1 by eliminating gradient conflicts.

Thank You!

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